

JOSEPH TANG

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EDUCATION

School of Information Science and Engineering, Southeast University

Bachelor of Information Engineering

Expected September 2027

Cumulative GPA: 3.89/4.00 | Average Score: 90.9

Relevant Coursework: Electromagnetic Fields and Waves (A), Fundamentals of Analog Electronic Circuits (A), Comprehensive Course Design of Computer Science (A), Fundamentals of Circuit (A+), Signals and Systems (B+)

Awards: “Zhishan” Students’ Academic & Research Scholarship (2025)

Department of Electrical Engineering and Computer Sciences, University of California, Berkeley

Exchange Program

Aug.2025 – Dec.2025

Relevant Coursework: Great Ideas of Computer Architecture (Machine Structures) (A), Introduction to Artificial Intelligence (A), Information Theory and Coding (B+)

TECHNICAL SKILLS

Software: CST Studio Suite, VS Code, MATLAB, Git Bash, Docker, Keil uvision5, Multisim, Altium Designer10, Origin

Programming Languages: Python, C/C++, RISC-V, Assembly Language

English Fluency: IELTS 8.0(Sep.2026) | GRE 329(Sep.2024) | DET 130(Feb.2025)

RESEARCH INTERESTS

- Integrate Machine Learning(ML) with electromagnetic device and VLSI design.
- Develop LLM-based Human-Computer Interaction (HCI) devices and Embodied Intelligence.

RESEARCH EXPERIENCE

Research Assistant

Liwei Lin Lab, University of California, Berkeley, United States | Supervisor: Dr. Zihan Wang

Aug.2025 – Dec.2025

Ultrasound Sonochemistry by Weakening or Breaking Chemical Bonds

- Fabricate a piezoelectric PMUT for dual-mode ultrasonic transmission and reception.
- Implement Python-based dynamic beamforming on a TI TX7516 platform to enhance localized acoustic pressure and trigger cavitation.
- Investigate non-thermal drug release enabled by the implosion energy of cavitation bubbles, and experimentally validated the sonochemical catalysis mechanism.
- Verify through terephthalic acid fluorescence spectroscopy that hydroxyl radical generation increased by 10.5× over the control group under optimized conditions, with peak fluorescence intensity reaching 400 a.u.

Near-Field Focusing via Multi-Dimensional Control Based on AI-Assisted Fast Synthesis

Purple Mountain Laboratories, China | Advisor: Assoc. Prof. Fan Wu

Mar.2026

IEEE MTT-S Scholarship Application Project

- Lead an AI-assisted near-field focusing framework for antenna arrays with enhanced polarization controllability.
- Enable joint amplitude-phase-polarization control by introducing element polarization orientation as an additional DoF.
- Build CST-based training datasets and train a surrogate model to enable fast prediction of complex electric-field distributions in target near-field regions by designing a multi-objective optimization function.

Metal Fatigue Crack and Corrosion NDE Based on Spoof Surface Plasmon Polaritons
State Key Laboratory of Millimeter Waves, China | Advisor: Prof. Chao Liu

Sep.2024 – Mar.2026

- Use CST combined with MLP to design an effective coupling structure that efficiently converts PWs into SSPPs.
- Design a two-dimensional focusing lens to concentrate electromagnetic energy, achieving a 76% energy conversion efficiency, approximately 1.67× higher than previous structures.
- Design a spoof LSP resonator at the focus and optimize coupling distance until a clear resonance peak was observed.
- Leverage localized field enhancement to detect surface cracks, achieving a spatial resolution of 0.2 mm, and evaluated the sensitivity of the system under lift-off conditions.

Research Assistant
CUHK, Shenzhen | Supervisors: Dr. Kaiming Shen & Prof. Zhiquan Luo

EUSC × AISE Summer Camp, Research Track
Jul.5 – Jul.31, 2026

- Extended Blind Configuration Method (BCM)’s active-sensing model from plane to exact spherical waves for near-field localization by proposing a joint coherent matched-filter estimator.
- Collected data from programmable metasurfaces, synchronizing random 1-bit/2-bit configurations and received-power measurements under different transmitter positions and panel arrangements.
- Reduced 2D localization RMSE from 1.81 m to 2.0 mm and 3D RMSE from 2.79 m to 2.05 cm across experimental datasets, while preventing a jeopardy caused by phase-gradient sign ambiguity.

Multi-source Gesture Recognition Technology for Intelligent Control Applications Based on Machine Learning
Purple Mountain Laboratories, China | Advisor: Prof. Gang Xu

Sep.2024 – Jun.2025

- Combine a millimeter-wave radar with specialized camera and synchronize their spatiotemporal relationship.
- Extract hand-motion data using a TI AWR1843 radar sensor and decompose into TF, RA, and RD maps based on FFT.
- Developed a lightweight visual-processing pipeline based on MediaPipe to extract and track the spatial topological features of hand keypoints in real time within the ROI.
- Applied multimodal data fusion algorithms to combine visual semantic information with radar feature maps, achieving 87% recognition accuracy across 10 types of dynamic gestures.

INTERNSHIP

Data Analyst Intern
Appen, Shanghai, China

Jen.2026 – Feb.2026

- Contributed to SWE-Bench, a standardized benchmark used to evaluate LLMs on software engineering tasks.
- Curated and validated Issue-Pull Request pairs on GitHub to ensure high-quality datasets for model training.
- Utilized Python, Docker, and Git for data acquisition, processing, verification and environment management.

STUDENT AFFAIRS

Student Union Member

Sep.2023 – Sep.2024

- Involved in publicity design, organization, and execution for multiple large-scale events.
- Executed layout design and content publishing for University Student Union’s official account.
- Contributed to campus merchandise design and production of branded university souvenir series.

Student Volunteer

- Completed 104.8 volunteer hours across diverse initiatives including: library shelving assistance, student sports meeting supply, and talent showcases coordination.